



Reconstruction of missing features based on a low-rank assumption for robust speaker identification

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Outline

Overview of missing features recovery

Overview of speaker identification

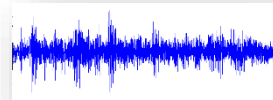
Low-rank missing data imputation

Experimental
Results

Conclusions &
Perspectives

Overview of missing features recovery

- ▶ Speech under noisy conditions
 - ▶ Difficult to achieve high recognition rates (ASR, speaker identification/verification, etc.)

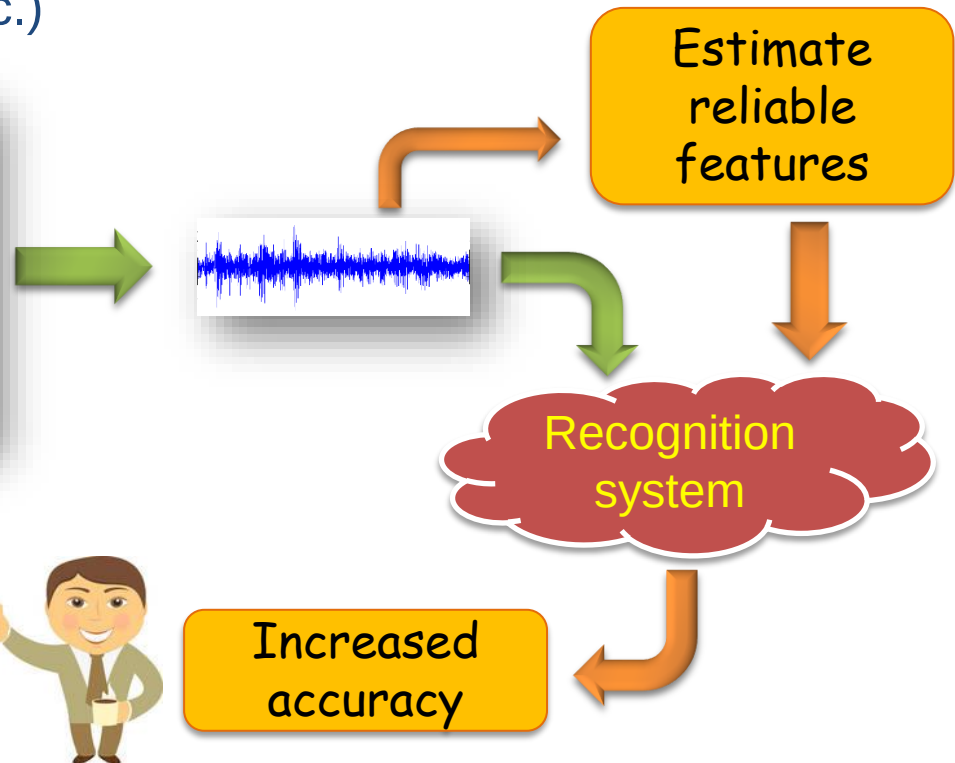


Low accuracy



Overview of missing features recovery

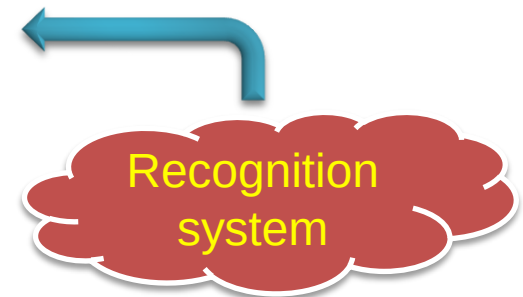
- ▶ Speech under noisy conditions
 - ▶ Difficult to achieve high recognition rates (ASR, speaker identification/verification, etc.)



Overview of missing features recovery

- ▶ Speech under noisy conditions
 - ▶ Difficult to achieve high recognition rates (ASR, speaker identification/verification, etc.)

Goal → *estimate speaker identity*



Overview of missing features recovery

- ▶ Speech under noisy conditions
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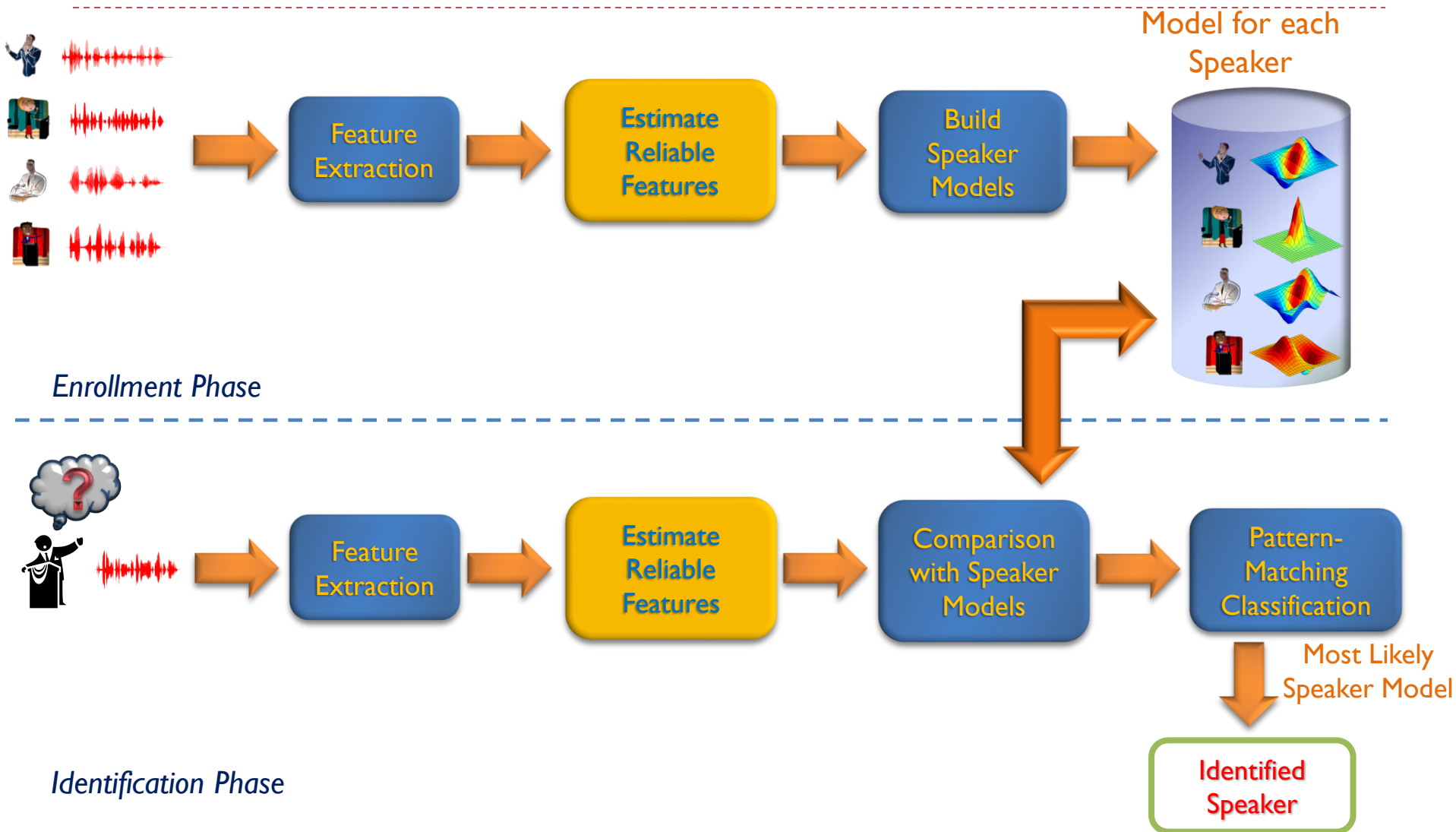
Goal → ***estimate speaker identity***



- Focus on current work
 - Text-independent
 - Short training and testing speech data
 - Highly noisy environments



Overview of speaker identification



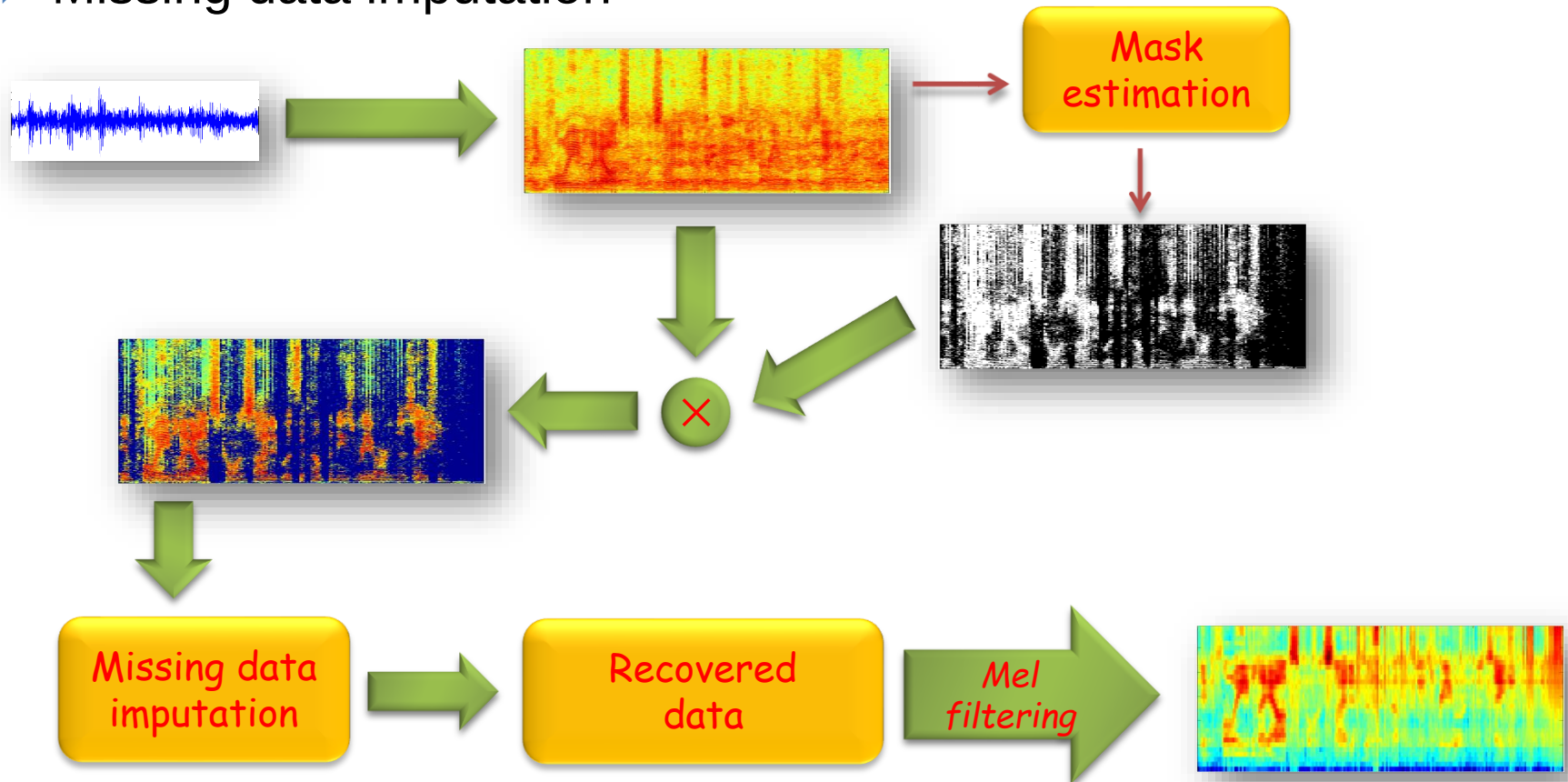
Overview of missing features recovery

- ▶ Noisy speech decomposed into
 - ▶ Speech-dominated components → reliable
 - ▶ Noise-dominated components → unreliable or missing
- ▶ Remove unreliable components → **Missing data regions**
 - ▶ Noisy spectrotemporal data
- ▶ Recover missing data adopting imputation approach
 - ▶ Produce reliable estimates of the missing spectrographic regions
- ▶ Reliability mask distinguishes reliable from unreliable (missing) data
- ▶ A lot of works dealing with practical reliability mask estimation

Focus → **missing data imputation**

Overview of missing features recovery

▶ Missing data imputation



Overview of missing features recovery

► Reliability mask estimation

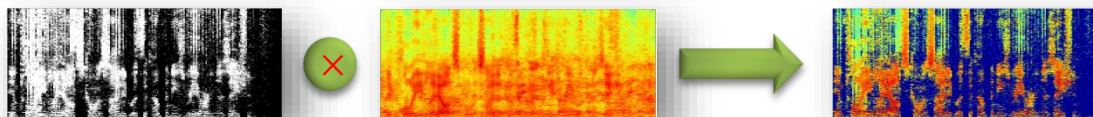
► Apply oracle binary mask

- observed spectrotemporal speech data $Y(f, \rho) = S(f, \rho) + N(f, \rho)$, where $f = 1, \dots, F$ and $\rho = 1, \dots, P$
 - $F \rightarrow$ FFT size
 - $P \rightarrow$ number of time frames
- $Y \in \mathbb{R}^{F \times P} \rightarrow$ log-magnitude STFT of the noisy speech signal
- $S \in \mathbb{R}^{F \times P} \rightarrow$ log-magnitude STFT of the clean speech signal
- $N \in \mathbb{R}^{F \times P} \rightarrow$ log-magnitude STFT of the contaminating noise

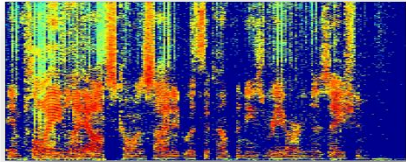
► Oracle binary mask estimation

$$\{0, 1\}^{F \times P} \ni W(f, \rho) = \begin{cases} 1 := \text{reliable}, & 10 \log_{10} \left(\frac{|S(f, \rho)|}{|N(f, \rho)|} \right) > \lambda \\ 0 := \text{unreliable}, & \text{otherwise} \end{cases}$$

SNR threshold λ

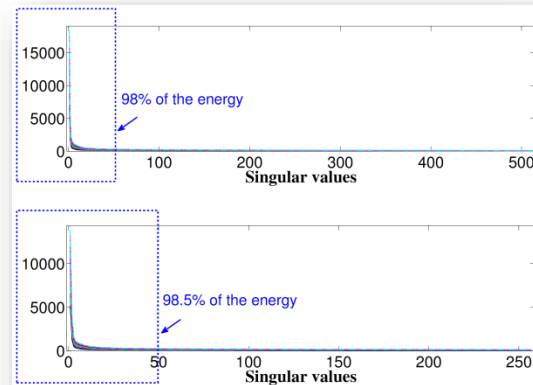
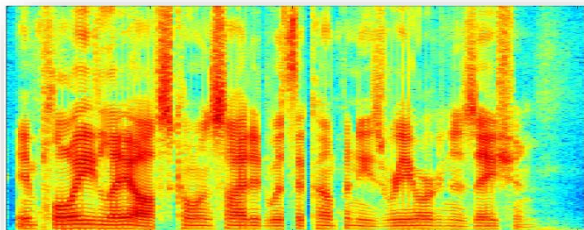


Low-rank missing data imputation

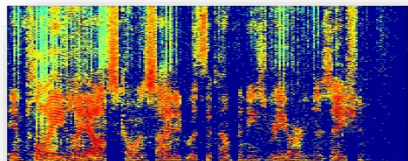


Could we complete missing data?

- ▶ **YES!** ... this is what *matrix completion* (MC) does
- ▶ **Core concept:** exploit *low-rank* behavior of speech features
- ▶ Speech spectrographic representation satisfies low-rankness? ✓



Low-rank missing data imputation



Could we complete missing data?

- ▶ **YES!** ... this is what **matrix completion** (MC) does
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Assumptions

- ▶ Observed entries of data matrix $M \in \mathbb{R}^{n_1 \times n_2} \rightarrow \mathcal{O}(nr \ln^2 n)$
- ▶ Known entries $M_{ij} \rightarrow (i, j) \in \mathcal{I} \subset \{1, \dots, n_1\} \times \{1, \dots, n_2\}$

$rank(M)$

$\max\{n_1, n_2\}$

Solve convex
opt. problem



$\min_{\mathbf{X}}$

$\|\mathbf{X}\|_*$

s.t. $X_{ij} = M_{ij}, (i, j) \in \mathcal{I} \subset \{1, \dots, n_1\} \times \{1, \dots, n_2\}$

$$\sum_{q=1}^{\min\{n_1, n_2\}} \sigma_q \text{ with } \sigma_1, \dots, \sigma_{\min\{n_1, n_2\}} \geq 0$$

Low-rank missing data imputation

- ▶ Define **linear map** $\mathcal{A} : \mathbb{R}^{n_1 \times n_2} \rightarrow \mathbb{R}^k \longrightarrow$ sampling operator recording a specific number of matrix entries ($k = |\mathcal{I}|$)
 - ▶ Use linear map to represent the constraints $X_{ij} = M_{ij}, \forall (i, j) \in \mathcal{I}$ as

$$\min_{\mathbf{X}} \|\mathbf{X}\|_* \quad \text{s.t. } \mathcal{A}_{\mathcal{I}}(\mathbf{X}) = \mathbf{b}$$

- $\mathbf{b} := \mathcal{A}_{\mathcal{I}}(\mathbf{M})$ contains the sample values extracted from $\mathbf{M}$
- $\mathcal{A}_{\mathcal{I}}(\mathbf{X}) \equiv \mathbf{A}\mathbf{x}, \mathbf{x} := \text{vec}(\mathbf{X}) \quad \forall \mathbf{X} \in \mathbb{R}^{n_1 \times n_2}$
 - $\mathbf{A} \in \mathbb{R}^{k \times n_1 n_2}$
 - $\text{vec}(\cdot) : \mathbb{R}^{n_1 \times n_2} \rightarrow \mathbb{R}^{n_1 n_2 \times 1}$
- Adjoint operator $\mathcal{A}_{\mathcal{I}}^*(\cdot) : \mathbb{R}^{k \times 1} \rightarrow \mathbb{R}^{n_1 \times n_2}$ with $k = |\mathcal{I}| \leq n_1 n_2$
 - Maps a vector into a sparse matrix
 - $\mathbf{h} = \mathcal{A}_{\mathcal{I}}(\mathcal{A}_{\mathcal{I}}^*(\mathbf{h})) \quad \forall \mathbf{h} \in \mathbb{R}^{k \times 1}$

Solve using **singular value thresholding (SVT)**



Low-rank missing data imputation

Solve using **singular value thresholding (SVT)**

$$\min_{\mathbf{X}} \tau \|\mathbf{X}\|_* + \frac{1}{2} \|\mathbf{X}\|_F^2 \quad \text{s.t. } \mathcal{A}_{\mathcal{I}}(\mathbf{X}) = \mathbf{b}$$

► Iterative process leads to solution

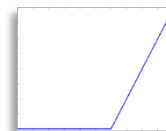
Algorithm 1: SVT algorithm

Input: $\mathcal{A}_{\mathcal{I}}$, observed values $\mathbf{b}$, step size δ , tolerance ϵ , parameter $\tau > 0$, maximum iterations $t_{\max}$
Output: estimated matrix $\mathbf{X}$

```
1 Initialization:  $\mathbf{y}_1 = \tau \mathcal{A}_{\mathcal{I}}^*(\mathbf{b}) / \|\mathcal{A}_{\mathcal{I}}^*(\mathbf{b})\|$ 
2 for  $t = 1$  to  $t_{\max}$  do
3    $[\mathbf{U}_t, \mathbf{\Sigma}_t, \mathbf{V}_t, s_t] = \text{SVDshrink}(\mathcal{A}_{\mathcal{I}}^*(\mathbf{y}_t), \tau, s_{t-1})$ 
4    $\mathbf{X}_t = \mathbf{U}_t \mathcal{D}_{\tau}(\mathbf{\Sigma}_t) \mathbf{V}_t^T$ 
5   if  $\|\mathcal{A}_{\mathcal{I}}(\mathbf{X}_t) - \mathbf{b}\|_2 \leq \epsilon \|\mathbf{b}\|_2$  then
6     break
7   end
8    $\mathbf{y}_{t+1} = \mathbf{y}_t - \delta(\mathcal{A}_{\mathcal{I}}(\mathbf{X}_t) - \mathbf{b})$ 
9 end
```

$$\begin{cases} \mathbf{X}_t = \mathcal{D}_{\tau}(\mathcal{A}_{\mathcal{I}}^*(\mathbf{y}_{t-1})) \\ \mathbf{y}_t = \mathbf{y}_{t-1} - \delta(\mathcal{A}_{\mathcal{I}}(\mathbf{X}_t) - \mathbf{b}) \end{cases}$$

- **Soft-thresholding** operator $\mathcal{D}_{\tau} = \mathbf{U} \mathbf{\Sigma}_{\tau} \mathbf{V}^T$
 - $\mathbf{\Sigma}_{\tau} = \text{diag}(\max\{\sigma_i - \tau, 0\})$
 - Singular values $\{\sigma_i\}_{i=1}^{\min\{n_1, n_2\}}$
- **Step size** δ



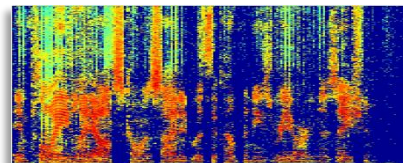
Low-rank missing data imputation

Missing-feature recovery using low-rank matrix completion

- Recover the missing spectrotemporal data $W \odot Y$

$$\hat{Y} = \arg \min_{\mathbf{X}} \|\mathbf{X}\|_* \quad \text{s.t.} \quad \mathcal{A}_{\mathcal{I}}(\mathbf{X}) = \mathcal{A}_{\mathcal{I}}(\mathbf{W} \odot \mathbf{Y})$$

$$\mathcal{I} = \{(i, j) \mid W(i, j) \neq 0\}, \quad \forall (i, j) \in \{1, \dots, F\} \times \{1, \dots, P\}$$



Compute mel-spectrogram



$$\left[\begin{array}{l} Q = 10 \cdot \log_{10} (B \cdot 10^{\hat{Y}/10}) \in \mathbb{R}^{d \times P} \\ \bullet \text{ mel-spaced filterbank amplitudes } B \in \mathbb{R}^{d \times F} \end{array} \right]$$



$$\left[\begin{array}{l} D = \Psi Q \\ \bullet \text{ discrete cosine transform (DCT) } \Psi \in \mathbb{R}^{d \times d} \end{array} \right]$$

Compute MFCCs



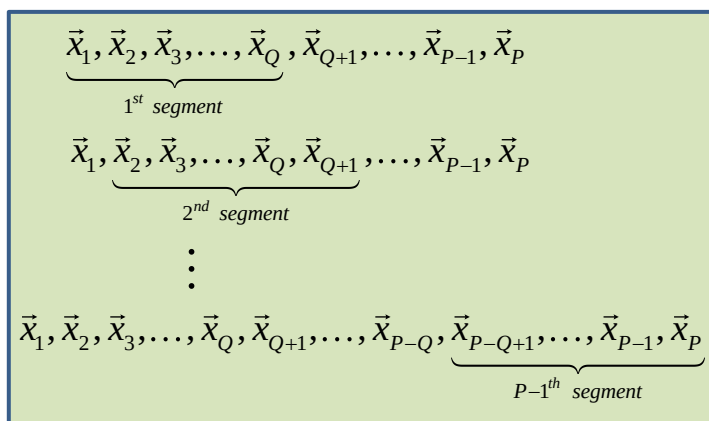
Experimental results

- ▶ Examine the efficiency of SVT in data recovery
 - ▶ Apply in robust speaker identification
- ▶ Compare with
 - ▶ Low-rank matrix fitting (LMaFit)
 - ▶ Scaled gradients on Grassmann manifolds (ScGrassMC)
- ▶ Speech corpus
 - ▶ VOICES → 12 speakers
 - ▶ Original F_s @ 22kHz → downsampling @ 16kHz
- ▶ Feature extraction
 - ▶ Analysis window → 40 msec (50% overlapping)
 - ▶ 30th order mel-frequency spectrogram

Experimental results

- ▶ Classification method ➡ UBM-GMM
 - ▶ 16 Gaussian mixtures
 - ▶ Diagonal covariance matrix
- ▶ Training data (clean speech) ➡ 10 sec per speaker
- ▶ Testing data ➡ segment length 400 (~8 sec)
 - ▶ Add noise during testing @ -15, -10, -5, 0, 5, 10 dB
 - ▶ Speech babble
 - ▶ Factory floor

NOISEX database



- Correct identification rate (CIR) of the j-th speaker

$$CIR_j = \frac{\# \text{ cor. identified segments}}{\text{total \# of segments}} \cdot 100\%$$

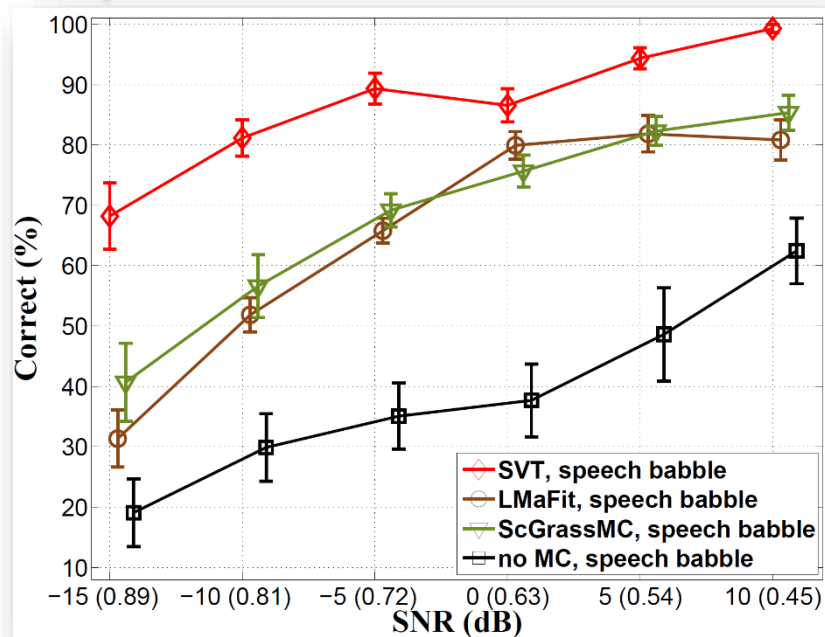
- Total mean correct identification rate

$$\text{mean CIR} = \frac{1}{R} \sum_{r=1}^R \left(\frac{1}{S} \sum_{s=1}^S CIR_s^r \right)$$

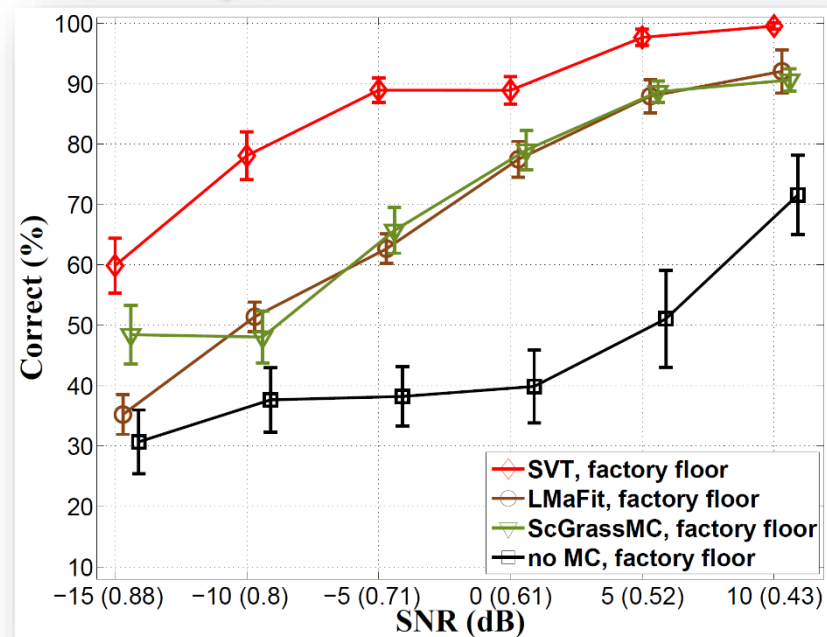


Experimental results

Correct identification rates (%) *speech babble*



factory floor



• numbers inside parentheses

$$1 - \text{Sampling ratio} = 1 - \frac{\text{number of observed values } (k)}{\text{matrix size } (F \times P)}$$

✓ SVT performs better

- Both noise types SNR @ -10 dB (~80% missing data) → mean CIR ~80%
- Both noise types SNR ≥ -5 dB → mean CIR ≥ ~87%



Experimental results

- ▶ Examine the efficiency of low-rank matrix recovery vs. sparse imputation
 - ▶ Sparse imputation (SI) based on sparsity assumptions
 - ▶ Supervised dictionary-based approach
- ▶ Add noise during testing @ -16, -12, -8, -4, 0, 4, 8, 12 dB
 - ▶ Speech babble
 - ▶ Factory floor

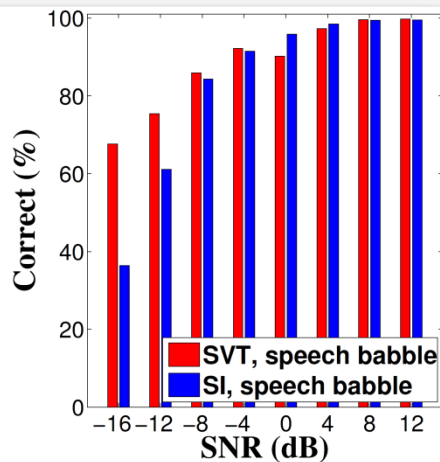
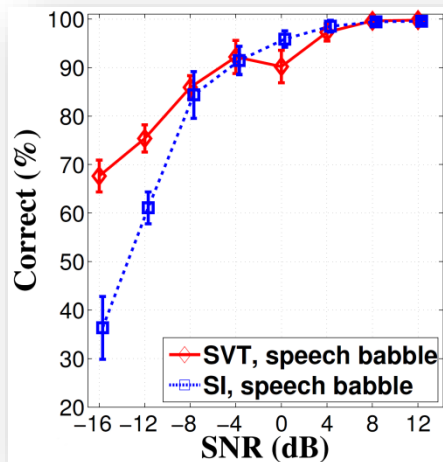
} *NOISEX database*



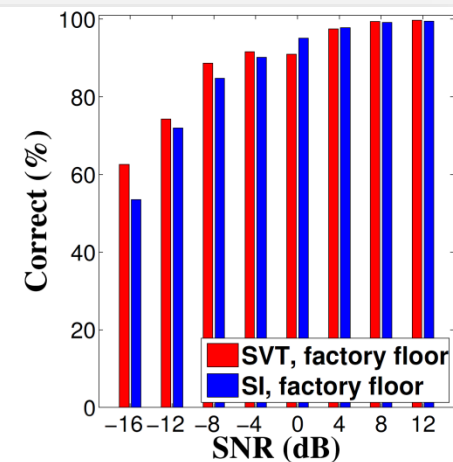
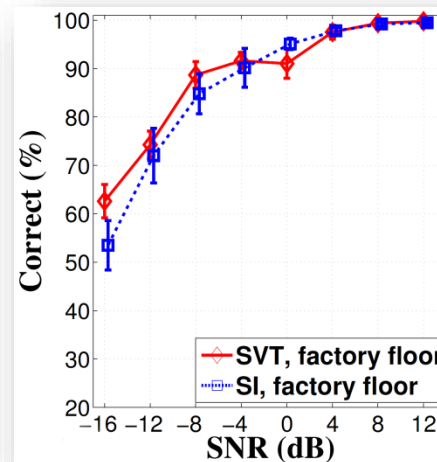
Experimental results

► Correct identification rates (%)

speech babble



factory floor



- ✓ SVT better vs. SI
 - Both noise types @ SNR ≤ -4 dB
- ✓ SVT 30% better vs. SI
 - Speech babble noise @ SNR = -16 dB
- ✓ SVT 15% better vs. SI
 - Speech babble noise @ SNR = -12 dB
- ✓ SVT 10% better vs. SI
 - Factory floor noise @ SNR = -16 dB



Conclusions & perspectives

- ▶ Enhance the reliability of noisy speech features for robust speaker identification
 - ▶ Adopt missing data imputation framework
- ▶ Exploit low-rankness of spectrographic speech data
 - ▶ Apply low-rank matrix completion to recover missing data regions
- ▶ Future perspectives
 - ▶ More simulations using practical reliability masks
 - ▶ Use SVT-based approach to produce reliable speech features and feed as input to other recognition systems e.g. ASR
 - ▶ Introduce a supervised version of SVT-based technique to enhance identification performance



Thank You!

